

Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$\mathcal{A}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{A}^{\bullet 1\bullet 2\bullet 3}$, {●1 → -a, ●2 → -b, ●3 → -c}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$\mathcal{A}_{0^{-}\alpha\beta\chi}^{\#1} - \frac{1}{2} \eta_{\beta\chi} \mathcal{A}_{1^{-}\alpha}^{\#1} + \frac{1}{2} \eta_{\alpha\chi} \mathcal{A}_{1^{-}\beta}^{\#1} + \frac{4}{3} \mathcal{A}_{2^{-}\alpha\beta\chi}^{\#1} + \frac{1}{3} \eta_{\beta\chi} \mathcal{A}_{0^{+}}^{\#1} n_{\alpha} + \mathcal{A}_{1^{+}\beta\chi}^{\#1} n_{\alpha} + \mathcal{A}_{2^{+}\beta\chi}^{\#1} n_{\alpha} -$ $\frac{1}{3} \eta_{\alpha\chi} \mathcal{A}_{0^{+}}^{\#1} n_{\beta} - \mathcal{A}_{1^{+}\alpha\chi}^{\#1} n_{\beta} - \mathcal{A}_{2^{+}\alpha\chi}^{\#1} n_{\beta} + \mathcal{A}_{1^{+}\alpha\beta}^{\#2} n_{\chi} - \frac{1}{2} \mathcal{A}_{1^{-}\beta}^{\#1} n_{\alpha} n_{\chi} - \mathcal{A}_{1^{-}\beta}^{\#2} n_{\alpha} n_{\chi} + \frac{1}{2} \mathcal{A}_{1^{-}\alpha}^{\#1} n_{\beta} n_{\chi} + \mathcal{A}_{1^{-}\alpha}^{\#2} n_{\beta} n_{\chi}$	$\sigma_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$\mathcal{A}_{0^{+}}^{\#1}$	Symmetry[0, $\mathcal{A}_{0^{+}}^{\#1}$, {}, StrongGenSet[{}, GenSet[]]]	$\mathcal{A}_{\alpha}^{\beta} n^{\alpha}$	$\sigma_{0^{+}}^{\#1}$
$\mathcal{A}_{0^{-}\alpha\beta\chi}^{\#1}$	Symmetry[3, $\mathcal{A}_{0^{-}}^{\#1\bullet 1\bullet 2\bullet 3}$, {●1 → -a, ●2 → -b, ●3 → -c}, StrongGenSet[{1, 2, 3}, GenSet[-(1,2), -(2,3)]]]	$\frac{1}{3} \mathcal{A}_{\alpha\beta\chi} - \frac{1}{3} \mathcal{A}_{\alpha\chi\beta} + \frac{1}{3} \mathcal{A}_{\beta\chi\alpha} - \frac{1}{3} \mathcal{A}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{3} \mathcal{A}_{\beta\delta\chi} n_{\alpha} n^{\delta} - \frac{1}{3} \mathcal{A}_{\chi\delta\beta} n_{\alpha} n^{\delta} +$ $\frac{1}{3} \mathcal{A}_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{A}_{\alpha\delta\chi} n_{\beta} n^{\delta} + \frac{1}{3} \mathcal{A}_{\chi\delta\alpha} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{A}_{\alpha\beta\delta} n_{\chi} n^{\delta} + \frac{1}{3} \mathcal{A}_{\alpha\delta\beta} n_{\chi} n^{\delta} - \frac{1}{3} \mathcal{A}_{\beta\delta\alpha} n_{\chi} n^{\delta}$	$\sigma_{0^{-}\alpha\beta\chi}^{\#1}$
$\mathcal{A}_{1^{+}\alpha\beta}^{\#1}$	Symmetry[2, $\mathcal{A}_{1^{+}}^{\#1\bullet 1\bullet 2}$, {●1 → -a, ●2 → -b}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$-\frac{1}{2} \mathcal{A}_{\alpha\chi\beta} n^{\chi} + \frac{1}{2} \mathcal{A}_{\beta\chi\alpha} n^{\chi} - \frac{1}{2} \mathcal{A}_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{A}_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta}$	$\sigma_{1^{+}\alpha\beta}^{\#1}$
$\mathcal{A}_{1^{+}\alpha\beta}^{\#2}$	Symmetry[2, $\mathcal{A}_{1^{+}}^{\#2\bullet 1\bullet 2}$, {●1 → -a, ●2 → -b}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$\mathcal{A}_{\alpha\beta\chi} n^{\chi} + \mathcal{A}_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} - \mathcal{A}_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta}$	$\sigma_{1^{+}\alpha\beta}^{\#2}$
$\mathcal{A}_{1^{-}\alpha}^{\#1}$	Symmetry[1, $\mathcal{A}_{1^{-}}^{\#1\bullet 1}$, {●1 → -a}, StrongGenSet[{}, GenSet[]]]	$-\mathcal{A}_{\alpha}^{\beta} n_{\beta} + \mathcal{A}_{\beta}^{\chi} n_{\chi} n_{\alpha} n^{\beta} + \mathcal{A}_{\alpha\beta\chi} n^{\beta} n^{\chi}$	$\sigma_{1^{-}\alpha}^{\#1}$
$\mathcal{A}_{1^{-}\alpha}^{\#2}$	Symmetry[1, $\mathcal{A}_{1^{-}}^{\#2\bullet 1}$, {●1 → -a}, StrongGenSet[{}, GenSet[]]]	$\mathcal{A}_{\alpha\beta\chi} n^{\beta} n^{\chi}$	$\sigma_{1^{-}\alpha}^{\#2}$
$\mathcal{A}_{2^{+}\alpha\beta}^{\#1}$	Symmetry[2, $\mathcal{A}_{2^{+}}^{\#1\bullet 1\bullet 2}$, {●1 → -a, ●2 → -b}, StrongGenSet[{1, 2}, GenSet[(1,2)]]]	$-\frac{1}{2} \mathcal{A}_{\alpha\chi\beta} n^{\chi} - \frac{1}{2} \mathcal{A}_{\beta\chi\alpha} n^{\chi} - \frac{1}{3} \eta_{\alpha\beta} \mathcal{A}_{\chi}^{\delta} n^{\chi} + \frac{1}{3} \mathcal{A}_{\chi}^{\delta} n_{\alpha} n_{\beta} n^{\chi} + \frac{1}{2} \mathcal{A}_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{A}_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta}$	$\sigma_{2^{+}\alpha\beta}^{\#1}$
$\mathcal{A}_{2^{-}\alpha\beta\chi}^{\#1}$	Symmetry[3, $\mathcal{A}_{2^{-}}^{\#1\bullet 1\bullet 2\bullet 3}$, {●1 → -a, ●2 → -b, ●3 → -c}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$\frac{1}{2} \mathcal{A}_{\alpha\beta\chi} + \frac{1}{4} \mathcal{A}_{\alpha\chi\beta} - \frac{3}{8} \eta_{\beta\chi} \mathcal{A}_{\alpha}^{\delta} n^{\delta} - \frac{1}{4} \mathcal{A}_{\beta\chi\alpha} + \frac{3}{8} \eta_{\alpha\chi} \mathcal{A}_{\beta}^{\delta} n^{\delta} - \frac{3}{8} \mathcal{A}_{\beta}^{\delta} n_{\alpha} n_{\chi} + \frac{3}{8} \mathcal{A}_{\alpha}^{\delta} n_{\beta} n_{\chi} + \frac{1}{4} \mathcal{A}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{2} \mathcal{A}_{\beta\delta\chi} n_{\alpha} n^{\delta} +$ $\frac{1}{4} \mathcal{A}_{\chi\delta\beta} n_{\alpha} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{A}_{\delta}^{\epsilon} n_{\alpha} n^{\delta} - \frac{1}{4} \mathcal{A}_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{2} \mathcal{A}_{\alpha\delta\chi} n_{\beta} n^{\delta} - \frac{1}{4} \mathcal{A}_{\chi\delta\alpha} n_{\beta} n^{\delta} - \frac{3}{8} \eta_{\alpha\chi} \mathcal{A}_{\delta}^{\epsilon} n_{\beta} n^{\delta} - \frac{1}{2} \mathcal{A}_{\alpha\beta\delta} n_{\chi} n^{\delta} -$ $\frac{1}{4} \mathcal{A}_{\alpha\delta\beta} n_{\chi} n^{\delta} + \frac{1}{4} \mathcal{A}_{\beta\delta\alpha} n_{\chi} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{A}_{\alpha\delta\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \eta_{\alpha\chi} \mathcal{A}_{\beta\delta\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \mathcal{A}_{\beta\delta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \frac{3}{8} \mathcal{A}_{\alpha\delta\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon}$	$\sigma_{2^{-}\alpha\beta\chi}^{\#1}$